

Venue-based Interventions in an Agent-Based Model of Residential Segregation

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Plan

- Venues as an important element of residential segregation
- Analytical (sociology) framework
- ABM Design concepts
- I will try to convince you that my model is reasonable
- Simulation results
- What is next and public engagement

Template Slide One: Model Overview

Agent-based modeling enables systematic testing of spatial interventions on neighborhood sorting dynamics.

- Street networks form local neighborhood connectivity, while expressways function as barriers[cite: 1, 2].
- Venues provide agents with alternative utility anchors outside immediate residential blocks[cite: 1, 2].
- Agents evaluate location utility using a weighted combination of neighbor composition and venue accessibility[cite: 1, 2].

Template Slide Two: Findings and Directions

Simulations reveal critical divergences across geographic scales following amenity-based interventions[cite: 1].

- Localized venue placement alters meso-level cluster concentrations without changing macro indices[cite: 1].
- System-wide integrated placements fracture macro residential segregation enclaves[cite: 1].
- Interactive physicalizations translate simulation dynamics into tangible public explorations[cite: 1, 2].